

Assignment - 2

Page No. _____

Date _____

Ajanta

Q1 What are the different protocols observed at the following layers of the Protocol Stack?

→ (i) Application layer:

The protocols observed at the application layer are

- HTTP
- HTTPS
- DNS

→ (ii) Transport layer:-

The protocols observed at the Transport layer are:-

- TCP
- UDP

→ (iii) Network layer:

The protocol observed at the network layer is

- IP (IPv4)

• IPv6 may also be observed depending on the network traffic

→ Conclusion:

Thus, during packet capture, different protocols such as HTTP/HTTPS, DNS, TCP, UDP, IPv4 and IPv6 can be observed at different layers.

Que 2 What is the total amount of data being received for the following two cases?

→ (i) When you access <http://neverssl.com/>

The total amount of data received should be obtained from the Wireshark capture by selecting the packets belonging to the HTTP connection and collecting the received TCP payload / data

∴ Total data received = [Enter the value shown in your Wireshark capture] bytes

→ (ii) When you access <https://www.cornell.edu>

Similarly, the total amount of the received data should be calculated from the packets belonging to HTTPS/TLS connection

= Total data received: [Enter the value shown in your Wireshark Capture]

Ques Answer the following when you access <http://neverest.com/>

→ (i) How many HTTP GET requests do you observe? List down the GET requests

→ During the capture, HTTP GET request can be identified using the Wireshark display filter

`http.request.method == "GET"`

→ The GET request observed should be listed from the packet capture

→ Number of HTTP GET request - [Wireshark Capture]

GET request

(a) GET / HTTP/1.1

(b) Any additional GET request shown in the capture should also be listed

✓ The HTTP version can be seen in the HTTP request / response details in Wireshark

→ (iii) for each of the HTTP GET request find out:-

(a) The total number of TCP segments being received :- for each HTTP GET request, identify the corresponding TCP data packet / segments carrying the HTTP response.

HTTP GET Request	TCP Segment Received
GET /	[value from Wireshark]
GET [second request]	[value from Wireshark]

(ii) The total amount of data being receive in the corresponding HTTP response message

HTTP GET Request	Data Received
GET /	[value] bytes
GET [second request]	[value] bytes

Ques Identify TCP three-way handshake used to establish the connection with <http://naver.com/>.

→ TCP connection establish has three-way handshake

(a) SYN

Client → Server

TCP flag: SYN

This packet initiates the TCP connection

• Sequence Number: -

(b) SYN - ACK

Server → Client

TCP flags: SYN, ACK

The server acknowledges the client SYN and send its own SYN

- sequence number:-

(c) ACK :-

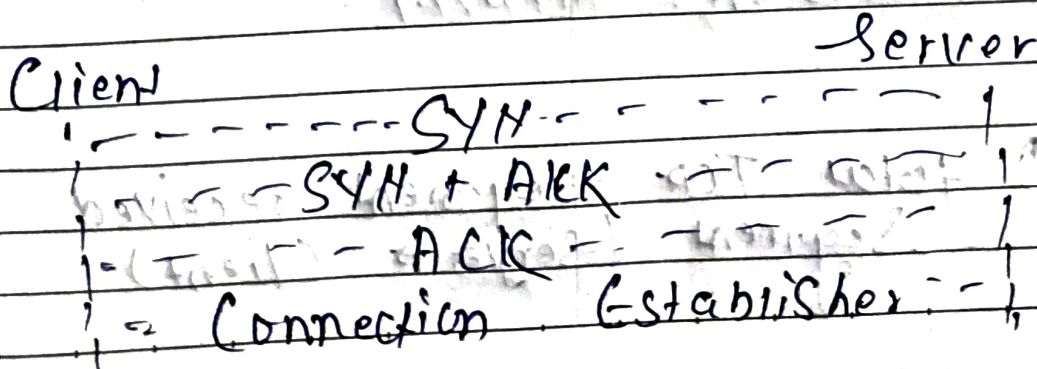
Client → Server

TCP flag: ACK :-

The client acknowledge the server's SYN and the TCP connection is established

- Sequence Number:-
- Acknowledgement No :-

→ Three-way handshake diagram



Conclusion:-

The TCP three-way handshake consists of:

SYN $\rightarrow$ SYN+ACK $\rightarrow$ ACK

$\rightarrow$ It is used to establish a reliable TCP connection b/w the client and the server.

3(i) Result

- packet No: 91176
- Request: GET / HTTP/1.1
- Host: never51.com
- TCP: Source port 53311 $\rightarrow$ Destination port 80
- TCP Seq: 1
- TCP len: 76 bytes
- From length: 130 bytes

3(ii) HTTP Version

$\rightarrow$ HTTP Version: HTTP/1.1

3(iii) (a) total TCP Segment received 2 TCP Segment (Server $\rightarrow$ Client)

(b) Total amount of data received in
HTTP response:
3961 bytes

3(iv) TCP 3 way handshake

1. SYN: Client 10.85.16.252 → Server
34.223.124.45. Sequence No = 0
2. SYN-ACK: Server 34.223.124.45 →
Client 10.85.16.252 Sequence No = 0
ACK No = 4
3. ACK: Client 10.85.16.252 → Server
34.223.124.45 Sequence No = 1
ACK No = 9